Lesson 5 Teacher's Checklist / Prep Guide

Complete before class

VIDEO GAME CREATION

Self-Contained Lesson Assets

No new assets this week - the first lesson where that's true. Everything needed is already sitting in students' own projects from Lessons 3 and 4: bullet_yellow.png, sfx_shoot.wav, enemy_invader_red.png, and sfx_hit.wav.

- ☐ No new Assets folder this week - confirm students still have their Lesson 3 and 4 Assets folders available if they need to re-import anything.

Continuity From Lesson 4

- ☐ Remind students to open their own [YourName]Game project from Lesson 4, not create a new one - today turns last week's single enemy into a full row of them.
- ☐ If a student's Lesson 4 project is lost or broken, have a pre-built "Lesson 4 complete" starter available per engine (from the Lesson 4 Teacher's Keys) so they aren't blocked.

Engine Verification

This week's core new concept - using a loop to create several instances instead of placing or spawning just one - has a genuinely different flavor per engine. Worth a quick personal test run before class.

- ☐ Confirm the demo laptop's Lesson 3 and 4 exercises still work for all 8 engines.
- ☐ Pre-build the full Scratch multi-clone exercise on top of a finished Lesson 4 project - this is the trickiest engine this week, since hit-detection has to move from the Bullet's script to the Enemy's own clone script (see the Key for why).
- ☐ Godot / Unity / GameMaker / Unreal: confirm a simple for-loop spawning several instances at different positions works cleanly in your installed version - standard, low-risk functionality, but worth a 60-second sanity check.
- ☐ GDevelop: confirm the "Repeat" event type (right-click an event sheet > Add Repeat Event) is available in your installed version.
- ☐ TIC-80: no new API here, just a for loop populating a Lua table at startup - low risk, but worth a quick test since the bullets-vs-enemies collision check now needs a nested loop instead of one comparison.
- ☐ 001 Game Creator (Windows demo laptop): confirm dragging several copies of the enemy Actor Template onto the Map each keep their own working "Collided with actor"

trigger with no extra scripting - standard Template behavior, but worth confirming once in your installed version.

- ☐ If any advanced-track students are on Unreal this week, confirm the Blueprint For Loop node is available and behaves as expected (Blueprints > Utilities > Flow Control).

Materials

- ☐ Print or queue digital copies of all 8 Student Handouts (or confirm students can access their engine's version electronically).
- ☐ Have the 8 Teacher's Keys on hand (instructor reference only - do not distribute) in case a student needs a worked example beyond Level 1.
- ☐ Bring: demo laptop, projector cable/adapter, handout copies.

Known Gotchas This Week

- Scratch: this is the one engine where the hit-detection logic actually has to move this week - Scratch clones have no unique ID, so a Bullet clone can't tell which specific Enemy clone it's touching. The fix (checking from each Enemy clone's own script instead) is the correct, standard Scratch technique - flag this clearly, since it's a real change from last week's code, not just an extension of it.
- Scratch (continued): because a hit Bullet clone is no longer explicitly deleted on contact, it keeps flying through a hidden/destroyed Enemy and cleans itself up normally off-screen - this is expected, not a bug, and worth explaining if a student asks why the bullet doesn't visibly vanish exactly on contact.
- Godot / Unity / GameMaker / GDevelop / Unreal: the most common slip is leaving the old, manually-placed single enemy in the scene/room/level alongside the newly spawned row - remind students to delete it once the spawn loop is working.
- 001 Game Creator: no scripting changes at all this week - just more copies of the same Actor Template placed on the Map. If a student expects to write a loop and finds nothing to write, that's correct, not a sign they're missing something.